

Incident Report Analysis

NIST CSF Analysis — ICMP Flood Denial of Service Attack

Summary	A multimedia company offering web design, graphic design, and social media marketing to small businesses experienced a denial of service (DoS) attack that compromised its internal network for approximately two hours. A malicious actor sent a flood of ICMP packets into the network through an unconfigured firewall, overwhelming network services so that normal internal traffic could not access any resources. The incident management team responded by blocking the incoming ICMP packets, taking non-critical services offline, and restoring critical services.
Identify	The attack was a denial of service (DoS) attack carried out as an ICMP flood, in which the attacker overwhelmed the network with a high volume of ICMP ping packets. The traffic entered through an unconfigured firewall, which was the vulnerability that made the attack possible. Systems affected: the internal network and its network services, which stopped responding and were inaccessible to internal users throughout the attack. Process affected: internal business operations that rely on network resources were interrupted for the two-hour duration of the outage. Impact: an estimated two-hour outage of internal network services.
Protect	To reduce the likelihood of a similar attack, the network security team put the following safeguards in place: Rate limiting: a new firewall rule that limits the rate of incoming ICMP packets, so a flood of pings can no longer overwhelm the network. Source IP verification: firewall verification of source IP addresses to detect and reject spoofed ICMP packets. IDS/IPS: an intrusion detection and prevention system that filters ICMP traffic based on suspicious characteristics. Firewall maintenance: regular review of the firewall configuration so that it is never left in an unconfigured state again.
Detect	To catch a similar attack earlier, the organization should continuously monitor network traffic for suspicious activity: Traffic monitoring: network monitoring software that flags abnormal traffic patterns, such as a sudden flood of ICMP packets from non-trusted sources. Log analysis: a SIEM tool to analyze network logs and alert the security team to unusual activity as it develops.
Respond	When a similar attack is detected, the response plan follows the steps the team used during this incident: Contain: block the malicious traffic at the firewall and take affected resources offline to limit the impact of the attack. Analyze: review network logs and traffic captures to confirm the attack type and trace its source.

	Communicate and improve: notify IT staff and affected users, and update the response procedures based on what the incident reveals.
Recover	Recovery returns the affected services to normal operation once the attack has been contained: Restore: bring critical network services back online first, then the non-critical services that were taken offline, once the network is stable. Improve: assess whether the two-hour recovery time can be shortened and refine the recovery process so future outages are resolved faster.

Reflections / Notes	
The root cause of this incident was a preventable misconfiguration: the firewall was left unconfigured, which gave the attacker a direct path into the internal network. The strongest single lesson is that basic firewall hardening (rate limiting, source IP verification, and filtering ICMP by suspicious characteristics) would likely have prevented the outage entirely. Running the incident through the five NIST CSF functions shows how a single event feeds a broader security strategy: Identify scopes what was hit, Protect and Detect reduce the chance and improve the visibility of a repeat, and Respond and Recover shorten the impact if it happens again.	